INTERNET-OF-THINGS AND EMBEDDED SYSTEM PRACTICE

Module Code: CSC83309

LECTURE

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Basic Arduino Programming Syntax and Explanation

1. Structure of an Arduino Sketch

An Arduino program is called a sketch and consists mainly of two functions:

```
void setup() {
  // Code here runs once when the Arduino starts
}
void loop() {
  // Code here runs repeatedly in a loop
}
```

- setup(): This function is called once at the beginning. Use it to initialize settings, pin modes, start serial communication, etc.

- loop(): This function runs continuously after setup(). Place your main code here.

2. Basic Commands

Pin Modes:

pinMode(pinNumber, INPUT);   // Set a pin as input

pinMode(pinNumber, OUTPUT); // Set a pin as output

Digital Write:

digitalWrite(pinNumber, HIGH); // Set pin to HIGH (5V)

digitalWrite(pinNumber, LOW);   // Set pin to LOW (0V)

Digital Read:

int value = digitalRead(pinNumber); // Read digital value (HIGH or LOW)

Delay:

delay(milliseconds); // Pause program for specified milliseconds

3. Example: Blink an LED

```
void setup() {
  pinMode(13, OUTPUT);  // Set pin 13 as output (built-in
LED)
}
void loop() {
  digitalWrite(13, HIGH);  // Turn LED ON
  delay(1000);             // Wait for 1 second (1000 ms)
  digitalWrite(13, LOW);   // Turn LED OFF
  delay(1000);             // Wait for 1 second
}
```

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Explanation:

- The LED connected to pin 13 will blink ON and OFF every second.
- pinMode configures the pin.
- digitalWrite changes the pin voltage.
- delay pauses the program.

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4. Variables

- int sensorValue = 0; // Declare an integer variable
- Use variables to store data like sensor readings.

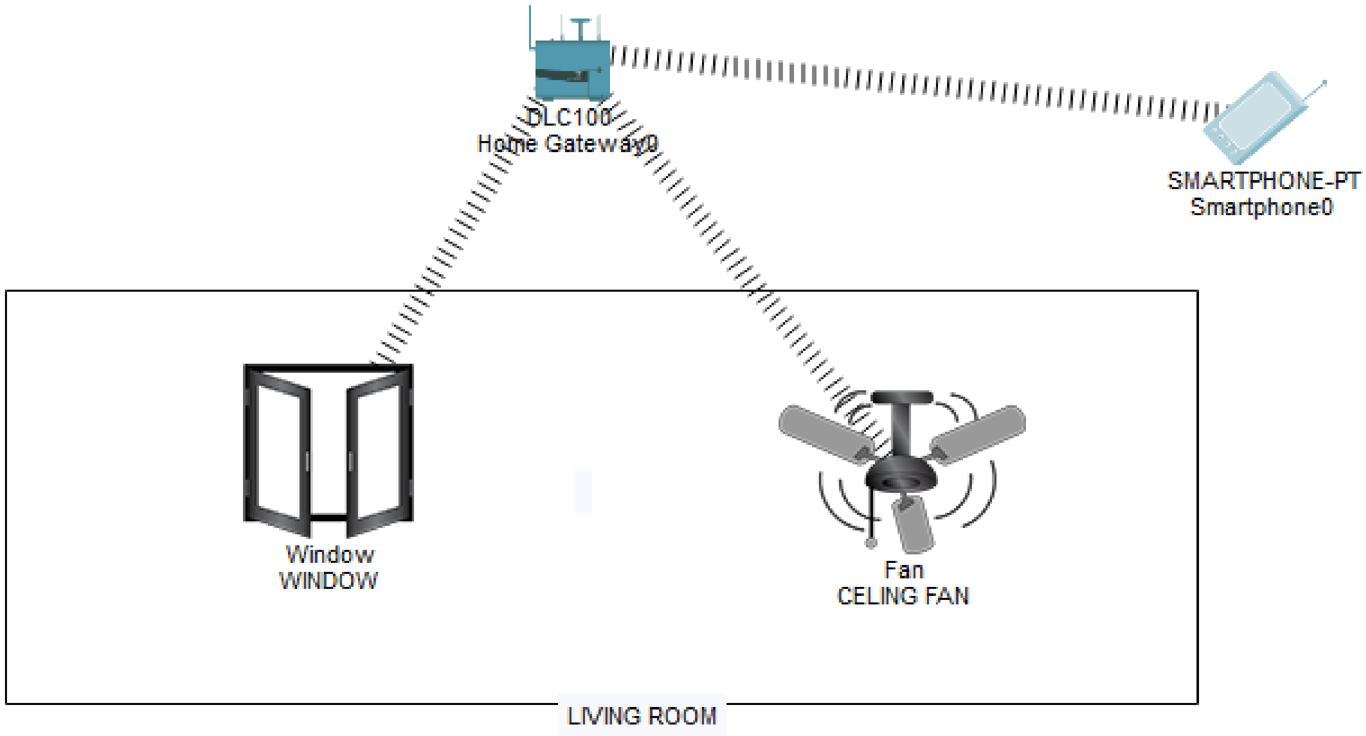
5. Comments

```
// This is a single-line comment

/*
 This is a
 multi-line comment
*/
```

Comments help explain the code and are ignored by the compiler.

Home automation using smart phone



2. Temperature monitoring